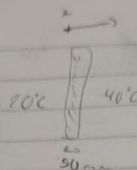


HW 2

Q. 1.6:



$$T(x=0) = 293 \text{ K}$$

$$T(x=0.05) = 313 \text{ K}$$

conduction

$$-k \frac{dT}{dx} = 40 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

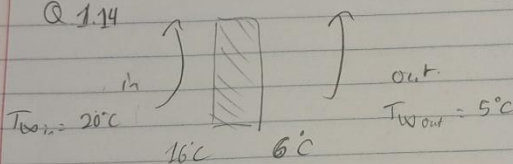
$$T(x) = \frac{313 - 293}{0.050} x + 293 \Rightarrow T(x) = 400x + 293$$

$$\frac{dT}{dx} = 400 \frac{\text{K}}{\text{m}} \Rightarrow k = 40 \frac{1}{dT/dx} = \frac{40}{400} = 0.10 \frac{\text{W}}{\text{m} \cdot \text{K}}$$

$$h_{\text{conv, in}} = 5 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$h_{\text{conv, out}} = 20 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

Q. 1.14



$$Q = mc \Delta T$$

$$\frac{dQ}{dt} = mc \Delta T \frac{dT}{dt}$$

$$\text{Convection: } J = h(T_{\text{in}} - T_{\text{out}})$$

$$J_{\text{conv, in}} = 5 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (16 - 20)^\circ\text{C} = -20 \frac{\text{W}}{\text{m}^2}$$

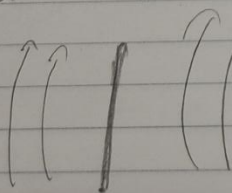
$$J_{\text{conv, out}} = 20 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (6^\circ\text{C} - 5^\circ\text{C}) = 20 \frac{\text{W}}{\text{m}^2}$$

The wall is at steady state

Consider this as 1-D wall, heat must flow from inner to outer surface with

$$\dot{Q}_{\text{conduction}} = 20 \frac{\text{W}}{\text{m}^2}$$

Q. 1.15



$$\text{Convection } \frac{dT}{dt} = -0.028 \frac{\text{K}}{\text{s}}$$

$$h = ? \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$T_{\text{in}} = 25^\circ\text{C}$$

$$T_{\text{out}} = 245^\circ\text{C}$$

$$\frac{J}{\text{m}^2 \cdot \text{K}}$$

$$m = 425 \text{ kg}$$

$$A = 0.4 \times 0.4 \times 2 = 0.32 \text{ m}^2 (1 \text{ face})$$

$$C = \frac{dT}{dt}$$

$$C = 2770 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$mc \frac{dT}{dt} = hA(T_{\text{in}} - T_{\text{out}})$$

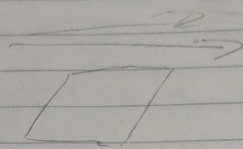
$$4.25 \times 2770 \times (-0.028) = h \times 0.32 \times (25 - 245) \Rightarrow h = 4.816 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$



②

Power max = ?

Convection



Q.1.13

$$h_1 = 200 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$h_2 = 3000 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$T_{\infty} = 15^\circ\text{C}$$

$$T_s = 85^\circ\text{C}$$

$$A = 25 \text{ mm}^2 = 25 \times 10^{-6} \text{ m}^2$$

$$P_{\text{max } 500} < h_{100} (T_s - T_{\infty}) A = 200 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} (85^\circ\text{C} - 15^\circ\text{C}) (25 \times 10^{-6}) \text{ m}^2 = 0.35 \text{ W}$$

$$P_{\text{max } 900} < h_{2000} (T_s - T_{\infty}) A = 3000 \times (85 - 15) (25 \times 10^{-6}) = 5.25 \text{ W}$$